

Changing choices is not yet a validated preference change

Question: can Sparse Autoencoder (SAE) steering produce a stable risk-choice policy across rewards, wording and answer labels?

ACTION ANCHOR

Raw logit gradient

A residual-space direction built from sensitivity of risky-versus-safe answer scores, averaged over discovery prompts. No SAE is used.

Observed: passed the action-control check at tested valid doses.

INTENDED PREFERENCE ANCHOR

Low-Rank Adaptation

LoRA is a trained weight-space adapter. We reused one trained to imitate a known Cumulative Prospect Theory (CPT) policy.

Observed: passed original-wording checks; failed rewording.

METHODS EVALUATED

Dense and SAE steering

A free residual direction versus a combination of ten SAE decoder features. Both aim at the same target policy, with matched edit size and positions.

Observed: neither method passed.

HELD-OUT RESULTS FOR THE COMBINED CPT TARGET

Error relative to the intended policy	Original wording	New wording
LoRA reference	0.042	0.115
Dense steering	0.129-0.151	0.224-0.225
Ten-feature SAE steering	0.202-0.208	0.220-0.224

Root Mean Squared Error (RMSE) of choice probabilities; lower is better. Declared accuracy limit: 0.060. Steering ranges show two training seeds, not confidence intervals.

Decision: defer the full feature sweep. The tested recipe failed. A robust positive anchor and a passing dense control are still missing, so this does not establish a general limitation of the SAE dictionary.

CPT gives us a known choice policy to aim for

CPT = Cumulative Prospect Theory. Here it is a mathematical behavioral benchmark, not a claim that a language model has human feelings.

Illustrative question: choose a guaranteed gain of 40 tokens, or a 50% chance of gaining 100 tokens and a 50% chance of gaining nothing. One answer cannot identify a preference; we vary outcomes and probabilities to measure a pattern.

1 / CURVATURE: ALPHA

Sensitivity to amounts

How does subjective value change as gains or losses grow? Our combined target uses **alpha = 0.8**, giving diminishing sensitivity in magnitude. The same exponent is used for gains and losses.

2 / WEIGHTING: GAMMA

Treatment of probabilities

Outcome probabilities are converted into decision weights using cumulative probabilities. Our combined target uses **gamma = 0.72**; gamma = 1 would leave probabilities undistorted.

3 / LOSS AVERSION: LAMBDA

Relative weight of losses

How strongly does a loss count relative to an equal-sized gain? Our combined target uses **lambda = 2**: the value magnitude of a loss is twice that of the corresponding gain.

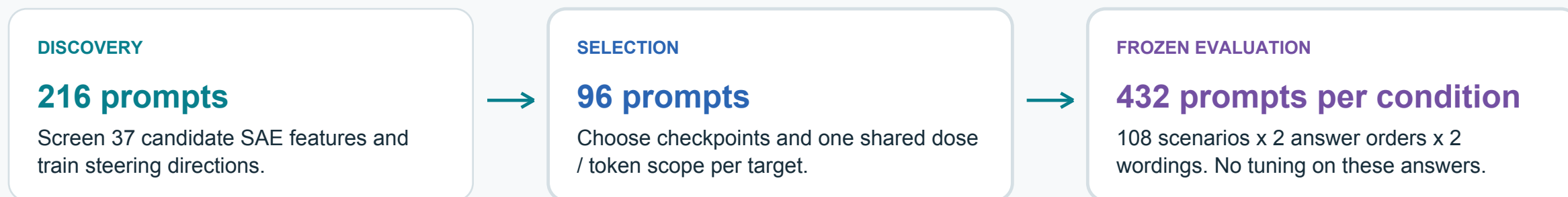
Two additional fitted quantities: choice sensitivity (inverse temperature, beta) controls how sharply value differences affect answer probabilities; an A-label bias captures a tendency to select the option displayed as A. Our CPT fit therefore has **five parameters**, not only three.

Two steering targets: combined = (alpha 0.8, gamma 0.72, lambda 2); neutral = (1, 1, 1). Both use beta = 3 and zero A-label bias. The neutral target is linear and probability-unweighted, but choices remain probabilistic.

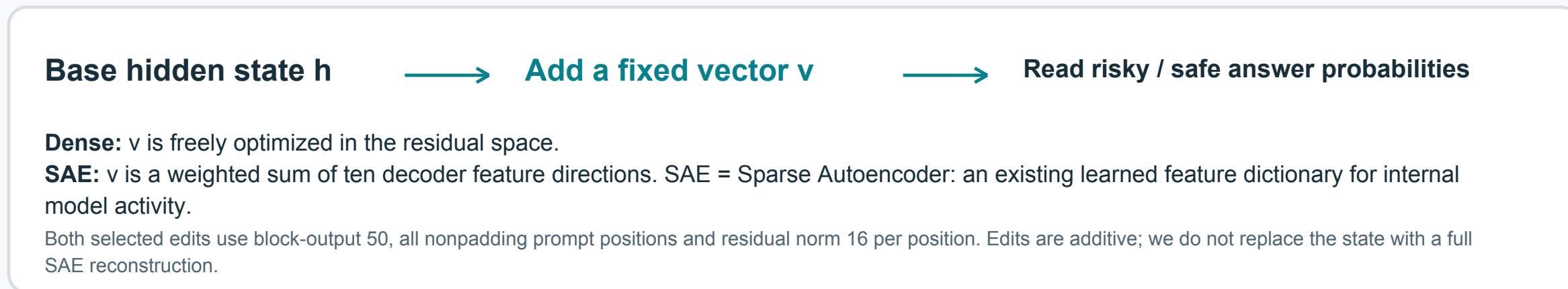
Implementation: outcomes are changes from a zero reference, scaled by 100 tokens. Gain/loss curvature and probability weighting are shared in this restricted CPT family. It does not cover every preference model. The example above is illustrative, not a measured test row.

Learn on development prompts; judge on untouched prompts

LLM = Large Language Model. We used the same Llama-3.3-70B-Instruct checkpoint in bfloat16 (bf16), a 16-bit numerical format.



WHAT WAS MATCHED BETWEEN DENSE AND SAE STEERING



What matching does and does not mean: the two methods receive the same-sized residual displacement at the same locations. That does not guarantee equal behavioral strength or optimization difficulty. The weight-space LoRA cannot be norm-matched this way.

Other controls: the unmodified model, reused LoRA, raw positive/negative readout gradients, sparse SAE readout approximation, random directions, and 48 elementary arithmetic/probability/dominance items. There were 28 final conditions in total.

Seeds are independent optimization initializations, not independent base models. Feature screening ranked full-grid choice loss, not direct switching-point / curvature effects. No new SAE or LoRA was trained in this latest round.

Why we used the phrase "bias-like"

This is a held-out predictive comparison between two explanations of the observed SAE-steered choices. It is not a semantic label for an SAE feature.

EXPLANATION A / 5 FITTED PARAMETERS

Restricted CPT model

Predict choices from rewards and probabilities using curvature, weighting, loss aversion, choice sensitivity and A-label bias.

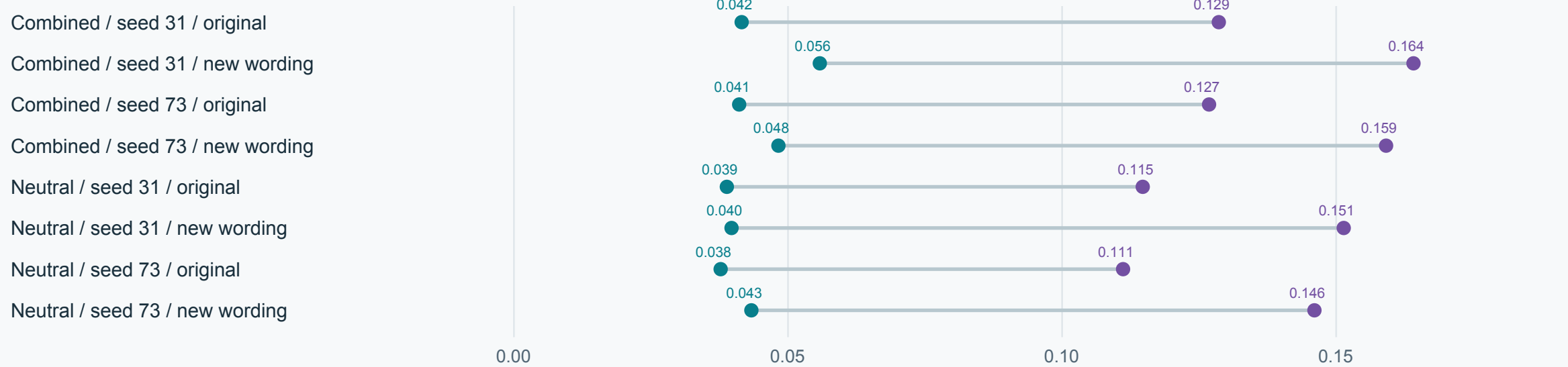
EXPLANATION B / 7 FITTED PARAMETERS

Answer-bias / confidence model

Transform the base model's risky/safe log-odds with a separate shift and positive scale for each of three frames, plus an A-label bias.

Both explanations were fitted on selection responses and then frozen. The chart compares their error when predicting the actual SAE-steered probabilities on unseen scenarios and wording.

HELD-OUT EXPLANATORY RMSE / LOWER IS BETTER



Interpret carefully: the bias model has more parameters and access to base-model responses. Better held-out prediction supports this descriptive explanation; it does not prove an internal mechanism or exclude every preference model.

The current recipe failed; the general SAE question stays open

Keep target recovery, behavioral model comparison, and feature meaning as three separate claims.

QUESTION 1 / DID THE INTERVENTIONS RECOVER THE PLANTED POLICY?

Target	Intervention	Original wording	New wording
combined	LoRA	0.0416	0.1147
combined	Dense vector	0.129-0.151	0.224-0.225
combined	Ten SAE features	0.202-0.208	0.220-0.224
neutral	Dense vector	0.1319-0.1323	0.241-0.248
neutral	Ten SAE features	0.2035-0.2043	0.2532-0.2533

RMSE against the planted teacher; target <= 0.060. Ranges span two seeds, not uncertainty intervals. These are different errors from page 4, whose target was the actual steered behavior.

ACTION REFERENCE

Raw gradient: pass

The valid-dose raw gradient met the operational action-control criterion. That validates one useful negative control.

PREFERENCE REFERENCE

LoRA: partial success

Passed the latest provisional criteria in original wording. Rewording failed recovery, prediction and answer-order checks.

STEERING OUTCOME

Dense 0/8; SAE 0/8

Eight combinations = 2 targets x 2 seeds x 2 wordings. None passed the complete rule; this is not eight separate tests.

Beyond exact target recovery: a separate preference-like check removed the planted-parameter and teacher-accuracy requirements. Both methods still passed 0/8. Yet failure of a finite candidate set and optimizer budget is not evidence that every SAE intervention must fail.

The earlier stricter repair rule used teacher RMSE <= 0.040; the reused LoRA misses it on this new grid in both wordings. "Pass" above refers only to the latest provisional rule. These thresholds do not replace the collaborator's full go/pivot protocol.

What to align with coauthors before another GPU run

The freely trained dense vector is a new diagnostic control. The original probe is also a dense residual direction, learned in a different way.

	Current dense control	Original Llama probe code
How learned	Optimize CPT choice-probability loss	Train a linear classifier on activations
Where applied	Block-output 50; selected all prompt tokens	Layer 48; final 20% with a position ramp
Comparison status	Evaluated in this pilot	Not rerun under these matched conditions

We cannot infer how the original probe would perform from this dense control's failure. A fair head-to-head needs aligned prompts, layer, token scope, dose, training objective and held-out criteria.

INTERPRETABILITY STATUS

Existing dictionary; our analyses

We used Goodfire's released layer-50 SAE. Descriptions are cached Neuronpedia labels, not independently validated meanings. Activations and steering effects are our measurements.

SPARSE READOUT DIAGNOSTIC

Large residual is scope-limited

The ten-feature readout approximation left relative residual norm 0.961. This diagnoses that sparse fit; it does not show that the full dictionary cannot express the gradient.

Next gate: validate a LoRA across new phrasings, label orders and reward grids. If it passes, repeat the small matched dense / SAE comparison. If dense succeeds and SAE fails, the sparse recipe is implicated more specifically. If both fail, change the intervention format before a full sweep.

Definitions and sources: CPT: Tversky & Kahneman (1992), [doi:10.1007/BF00122574](#). LoRA: Hu et al. (2021), [arXiv:2106.09685](#). SAE: [Goodfire layer-50 model card](#). Result source: decision-refined-20260918T235147Z, decision_analysis.json. Local source index, full methods and figure data accompany this brief.

Prepared with Codex using saved results checked against the analysis code. This brief adds no experimental data; its figures are drawn from the saved analysis. The previous GPU session was stopped and its artifacts verified.